KINGSCLIFF HIGH SCHOOL

ACHIEVEMENT THROUGH ENDEAVOUR

NAME:

TEACHER:

MATHEMATICS

Year 9 5.2

Yearly Exam Practice

Practice Exam 4

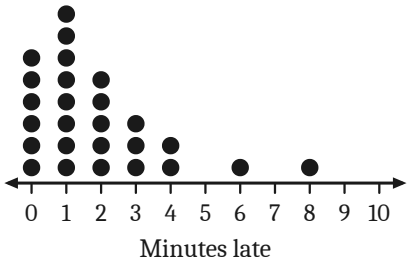
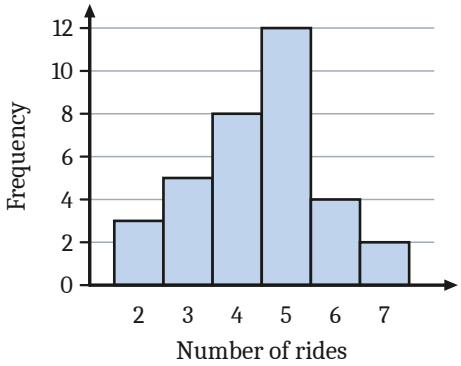
Instructions

1. Working time: 60 minutes
2. Attempt all questions
3. Show ALL working where needed
4. A calculator is permitted
5. Answers are at the back of this booklet

Sections and Marks

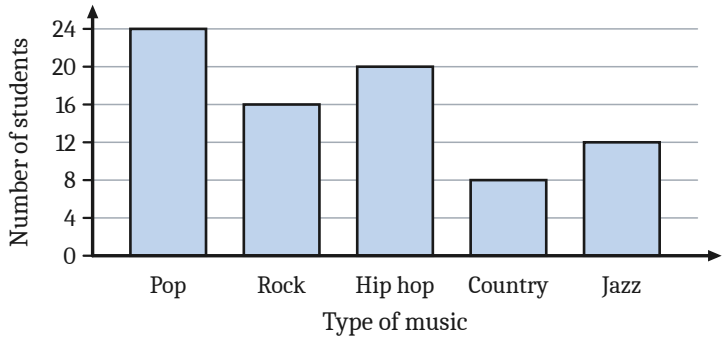
Section 1 - Data	/14
Section 2 - Surface area and volume	/20
Section 3 - Mixed Topics	/30
Total marks	**/64**

Section 1: Data

	Multiple Choice:	Mark
1	Which of the following is **numerical continuous** data? A. The number of cars in a car park B. The type of phone a student owns C. The number of people on a bus D. The volume of water in a bottle	1
2	The dot plot shows how many minutes late a train was on each of 26 school days. Which statement best describes the data? A. Negatively skewed, clustered at 0 and 1 B. Positively skewed, clustered at 0 and 1 C. Symmetrical, clustered at 0 and 1 D. Positively skewed, clustered at 6 and 8 	1
3	The histogram shows the number of rides each of 34 students went on at a show. What is the mode? A. 5 B. 7 C. 4.5 D. 12 	1
4	A shop recorded the number of customers each day for 8 days: 120, 125, 118, 122, 410, 121, 124, 122 Which measure of centre changes the most if the outlier, 410, is removed? A. The mode B. The mean C. The median D. None of them change	1

Section 1: Data (continued)

Multiple Choice:

		Mark
5	Students were asked about their favourite type of music. The column graph shows the results. What percentage of the students chose rock or country?  A. 30% B. 24% C. 20% D. 48%	1

Short Answer:

	Question	Working Out	Mark
6	The stem-and-leaf plot shows the number of text messages that 11 students sent in one day. Stem | Leaf 0 | 9 1 | 3 6 8 2 | 1 2 3 6 3 | 0 3 4 | 2 Key: 2 | 1 = 21 messages a. Find the mean. b. Find the median. Note: Mean = sum of the scores ÷ number of scores	a. b.	2 1

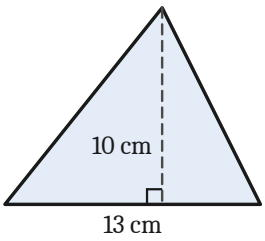
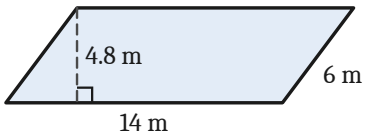
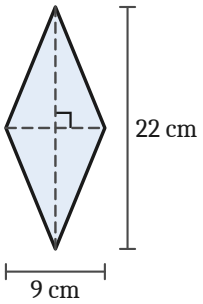
Section 1: Data (continued)

Worded Problems:

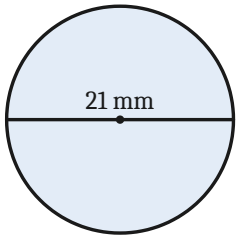
	Question	Working Out	Mark
7	The back-to-back stem-and-leaf plot shows the number of coffees sold each hour for 10 hours at two coffee shops. Note: Range = highest score − lowest score	Shop A | Stem | Shop B 0 | 8 8 7 4 | 1 | 3 6 9 7 5 4 2 2 1 0 | 2 | 4 7 9 3 | 1 5 8 Key: 2 | 4 = 24 coffees	
a	Find the median number of coffees sold for each shop.		2
b	Which shop usually sells more coffees? Give a reason.		1
c	Which shop is more consistent? Find the range of each shop to explain.		3
	Total		**/14**

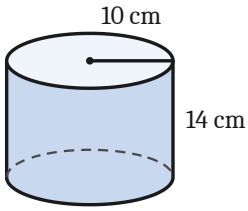
Section 2: Surface area and volume

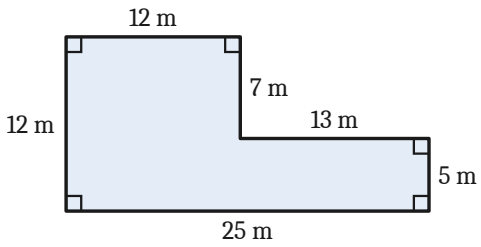
Multiple Choice:

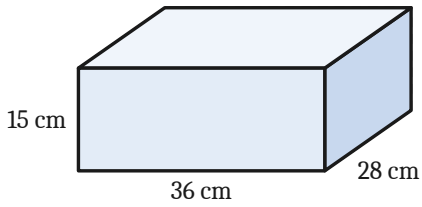
		Mark
1	A container has a volume of 6000 cm^3. How many litres does it hold? (1 L = 1000 cm^3) A. 60 L B. 600 L C. 0.6 L D. 6 L	1
2	What is the area of this triangle? A. 130 cm^2 B. 23 cm^2 C. 11.5 cm^2 D. 65 cm^2 	1
3	What is the area of this parallelogram? Note: Area of a parallelogram: $A = bh$ A. 84 m^2 B. 67.2 m^2 C. 33.6 m^2 D. 40 m^2 	1
4	What is the area of this rhombus? Note: Area of a rhombus: $A = \frac{1}{2}xy$ (x and y are the diagonals) A. 198 cm^2 B. 99 cm^2 C. 49.5 cm^2 D. 31 cm^2 	1

Short Answer:

	Question	Working Out	Mark
5	For this circle, find, correct to 1 decimal place:  a. its circumference b. its area. Note: Circumference: $C = 2\pi r$ Note: Area: $A = \pi r^2$	a. b.	1 1

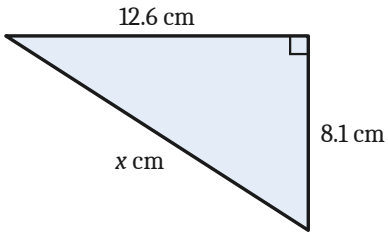
Section 2: Surface area and volume (continued)			
	Question	**Working Out**	**Mark**
6	A closed biscuit tin is a cylinder. It has a radius of 10 cm and a height of 14 cm.  a. Find its volume, correct to 1 decimal place. b. Find its surface area, correct to 1 decimal place. Note: Volume: $V = \pi r^2 h$ Note: Surface area: $SA = 2\pi r^2 + 2\pi rh$	a. b.	2 2

	Section 2: Surface area and volume (continued)		
	Worded Problems:		
	Question	**Working Out**	**Mark**
7	The plan shows an L-shaped concrete driveway and parking area. All of the corners are right angles. Note: Area of a rectangle: $A = lb$ 		
a	An edging strip goes around the outside of the concrete. How many metres of edging are needed?		1
b	Find the area of the driveway and parking area.		3
c	Concrete costs $55.20 per square metre. Find the cost of the concrete for the whole area.		1

	Section 2: Surface area and volume (continued)		
	Question	**Working Out**	**Mark**
8	A closed shoebox is shaped like a box. It is a rectangular prism 36 cm long, 28 cm wide and 15 cm high. 		
a	Find the volume of the box in cm^3. Note: Volume of a rectangular prism: $V = lwh$		1
b	How many litres does the box hold? Note: 1 L = 1000 cm^3		1
c	The outside of the shoebox is to be covered in wrapping paper. Find the surface area of the box. Note: Surface area = the total area of the 6 faces; area of a rectangle: $A = lb$		3
	Total		**/20**

Section 3: Mixed Topics (Chapters 1–8)

Multiple Choice:

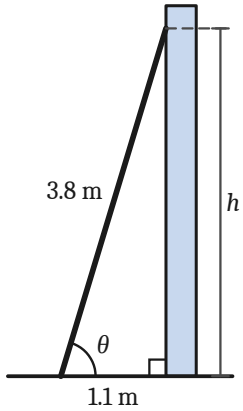
		Mark
1	If $x = 5$ and $y = 8$, evaluate $3x^2 + y$. A. 23 B. 38 C. 83 D. 233	1
2	Find the value of x. Select the closest answer. Note: Pythagoras' theorem: $c^2 = a^2 + b^2$ A. 20.7 B. 15.0 C. 9.7 D. 4.5 	1
3	A tent is marked at \$420. In a sale it is discounted by 35%. What is the selling price? A. \$567.00 B. \$385.00 C. \$273.00 D. \$147.00	1
4	Simplify $8w^5 \times 4w^4$. A. $32w^9$ B. $32w^{20}$ C. $12w^9$ D. $32w^1$	1
5	Ben, a police officer, earns an annual salary of \$85 280. How much is this per fortnight? Note: 1 year = 52 weeks = 26 fortnights = 12 months A. \$1640.00 B. \$3280.00 C. \$3553.33 D. \$7106.67	1

Short Answer:

	Question	Working Out	Mark
6	Simplify: a. $7a + 2b + 4a - 6b$ b. $36mn \div 9n$	a. b.	1 1
7	Expand and simplify: $3(2m + 5) - 5(m - 4)$		2

Section 3: Mixed Topics (Chapters 1–8) (continued)			
	Question	**Working Out**	**Mark**
8	Factorise: a. $9n - 24$ b. $w^2 + 11w$	a. b.	1 1
9	Mei invests \$12 500 for 5 years at 3.9% p.a. simple interest. Calculate the interest earned. Note: Simple interest: $I = Prn$		2
10	A furniture shop buys a couch for \$1250 and sells it for \$1450. a. Find the profit. b. Find the profit as a percentage of the cost price. Note: Percentage profit = profit ÷ cost price × 100%	a. b.	1 1
11	Solve: $10n - 3(n + 4) = 5n + 6$		3
12	Omar sells boats. Omar is paid \$550 per week plus 2% commission on sales. Last week Omar sold \$64 300 worth of boats. Calculate Omar's total pay for the week.		2

	Section 3: Mixed Topics (Chapters 1–8) (continued)		
	Worded Problems:		
	Question	**Working Out**	**Mark**
13	Isla works at a garden centre. The normal rate of pay is \$29.80 per hour for the first 38 hours each week, and time-and-a-half for each hour after that. Last week Isla worked 47 hours.		
a	Calculate the pay for the first 38 hours.		1
b	Calculate the total pay for the 47 hours.		2
14	A dog-walking business charges a \$30 sign-up fee plus \$18 per walk. The total cost, C dollars, for w walks is given by the formula $\boldsymbol{C = 30 + 18w}$		
a	Find the total cost for 6 walks.		1
b	Ella paid \$300 altogether. Write an equation and solve it to find how many walks Ella's dog had.		2

Section 3: Mixed Topics (Chapters 1–8) (continued)			
	Question	**Working Out**	**Mark**
15	A ladder 3.8 m long leans against the vertical wall of a shed. The foot of the ladder is 1.1 m from the wall, on level ground. Note: Pythagoras' theorem: $c^2 = a^2 + b^2$ Note: $\sin\theta = \frac{\text{opp}}{\text{hyp}}$, $\cos\theta = \frac{\text{adj}}{\text{hyp}}$, $\tan\theta = \frac{\text{opp}}{\text{adj}}$ 		
a	Find h, the height the ladder reaches up the wall, correct to 2 decimal places.		2
b	Find the angle θ that the ladder makes with the ground, correct to the nearest degree.		2
	Total		**/30**

Answers

Mark your own work. If an answer is wrong, read the help sheet topic in the last column, then try the question again.

Section 1 answers – Data			
Q	**Answer**	**Marks**	**If you got it wrong, read…**
1	D (The volume of water in a bottle)	1	D1 Types of data
2	B (Positively skewed, clustered at 0 and 1)	1	D2 Shape of a distribution
3	A (5)	1	D3 Reading graphs
4	B (The mean)	1	D5 Outliers
5	A (30%)	1	D3 Reading graphs
6a	23 messages	2	D4 Mean, median, mode, range
6b	22 messages	1	D4 Mean, median, mode, range
7a	Shop A: 21.5; Shop B: 25.5	2	D6 Comparing two data sets
7b	Shop B: the median for Shop B (25.5) is higher than the median for Shop A (21.5).	1	D6 Comparing two data sets
7c	Shop A: the range is 13 (27 – 14), much smaller than the range for Shop B, 30 (38 – 8).	3	D6 Comparing two data sets
	Total	**14**	

Section 2 answers – Surface area and volume			
Q	**Answer**	**Marks**	**If you got it wrong, read…**
1	D (6 L)	1	M1 Units
2	D (65 cm^2)	1	M2 Area of shapes
3	B (67.2 m^2)	1	M2 Area of shapes
4	B (99 cm^2)	1	M2 Area of shapes
5a	66.0 mm	1	M3 Circles
5b	346.4 mm^2	1	M3 Circles
6a	4398.2 cm^3	2	M5 Volume
6b	1508.0 cm^2	2	M6 Surface area
7a	74 m	1	M4 Composite shapes
7b	209 m^2	3	M4 Composite shapes
7c	$11 536.80	1	M4 Composite shapes
8a	15 120 cm^3	1	M5 Volume
8b	15.12 L	1	M1 Units
8c	3936 cm^2	3	M6 Surface area
	Total	**20**	

Section 3 answers – Mixed Topics (Chapters 1–8)			
Q	**Answer**	**Marks**	**If you got it wrong, read…**
1	C (83)	1	A1 Substitution
2	B (15.0)	1	T1 Pythagoras
3	C ($273.00)	1	F1 Percentages and profit
4	A ($32w^9$)	1	A5 Index laws
5	B ($3280.00)	1	F3 Earning money
6a	$11a - 4b$	1	A2 Simplifying
6b	$4m$	1	A2 Simplifying
7	$m + 35$	2	A3 Expanding
8a	$3(3n - 8)$	1	A4 Factorising
8b	$w(w + 11)$	1	A4 Factorising
9	$2437.50	2	F2 Simple interest
10a	$200	1	F1 Percentages and profit
10b	16%	1	F1 Percentages and profit
11	$n = 9$	3	E1 Solving equations
12	$1836	2	F3 Earning money
13a	$1132.40	1	F3 Earning money
13b	$1534.70	2	F3 Earning money
14a	$138	1	E2 Formulas and word problems
14b	15 walks	2	E2 Formulas and word problems
15a	3.64 m	2	T1 Pythagoras
15b	73°	2	T2 Trigonometry
	Total	**30**	